Solar Filament Segmentation Challenge 2026

Project overview

Sijan Regmi

What is a solar filament?

A solar filament is a long, thin, dark structure of cool, dense plasma suspended in the Sun's hot atmosphere (the corona) and held up by magnetic fields. Seen against the bright solar disk in H-alpha light (a red wavelength emitted by hydrogen), it looks like a dark thread. The same structure seen at the Sun's edge, glowing against dark space, is called a prominence.

Filaments matter for space weather. When the magnetic field holding one becomes unstable, the filament can erupt and help launch a coronal mass ejection, which can disturb satellites, communications and power grids on Earth. Finding and tracking filaments in images is therefore a first step toward studying and forecasting these events.

What does the challenge ask for?

The challenge, organized by the Earth-Space AI Research (ESAIR) Lab and hosted on Kaggle, asks participants to build an automatic method that finds filaments in full-disk H-alpha images from the GONG network (NSO Integrated Synoptic Program). The task is *instance segmentation*: for each image, output one pixel mask per filament, not just a count or a bounding box.

- **Data:** 707 labelled training images (8,199 manually drawn filament annotations, with some images annotated by more than one person) and 180 test images.
- **Output:** a CSV with one row per predicted filament: the image ID and the mask in run-length encoding.
- **Metric:** micro Panoptic Quality (PQ). A predicted filament counts as a match only when its intersection-over-union with an annotated filament is above 0.5. The score is the sum of matched IoUs divided by TP + $\frac{1}{2}$FP + $\frac{1}{2}$FN, pooled over all annotator records.
- **Rules:** any computer-vision method, with or without learning. Free, publicly available pretrained models are allowed. Participants also submit source code (a public Git repository) and a short technical report. The deadline is November 15, 2026.

This entry in brief

- A classical baseline (contrast filtering and connected components) scored 0.15 on the public leaderboard.
- A U-Net ensemble (ResNet34 and ResNet50 encoders, ImageNet-pretrained) predicts a probability map for each image.
- A learned component-tree selector then decides which candidate regions to keep as filaments.
- The final submission scores 0.38 on the public leaderboard (about half of the test data), with a local validation PQ of 0.44.

Code and the full technical report: `https://github.com/regmi-sijan/solar-filament-segmentation`